Fiscal Year:	FY 2022	**Task Last Updated:**	FY 12/05/2022
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Assessment of Nutritional Value and Growth Parameters of Space-Grown Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline-- Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Space Biology Unsolicited
Start Date:	08/01/2017	**End Date:**	10/31/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 10/31/2022 per NSSC information (Ed., 10/31/22) NOTE: End date changed to 2/28/2022 per NSSC information (Ed., 7/8/21) NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
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	The goal of the proposed work is to determine the effects of space flight on growth, performance (productivity and time to maturation), metabolism, and genetic changes in radish plants (Raphanus sativus). Plants will be grown in the ADVANCED PLANT HABITAT (APH) hardware on the International Space Station (ISS) and in comparable experiments in ground controls. The research conforms to NASA's top future goal of long-term space exploration and terraforming other planets and satellites. These goals depend on nutritionally valuable food crops that can be reliably grown in an environment that is not only defined by altered gravity or weightless environment but also different atmospheric conditions. In contrast to studying direct effects of gravity on plants, our approach focuses on secondary, derived conditions such as altered convection, transpiration, watering conditions, CO2, other gaseous components, and LED illumination, which creates a characteristic energy (heat and light) gradient within the growth volume. Radishes are good candidate plants to grow on the ISS because they mature quickly (harvest after = 28 days after planting), have been grown successfully on the ISS (Massa et al., 2013); genetic and metabolic information is readily available (Shen et al., 2013); and they are nutritionally desirable and provide a wealth of information that can thoroughly be analyzed by biochemical, physiological, and genetic approaches. Radishes are good sources of minerals (especially K, Ca, Fe, and Zn), complex carbohydrates, vitamin C, dietary fibers, folic		

Task Description:	acid (vitamin B9), anthocyanins, and glucosinolates, a class of secondary metabolites characteristic for Brassicaceae. In addition to their roots, certain varieties of radish are used for their leaves, siliques, and oil. The nutritional value and versatility of these vegetables makes them good candidates as a crop plant for astronauts. The small seed size (<4 mm), longevity of seeds (up to five years), small plant size, and the ability to consume the entire plant makes radishes ideally suited for space biology. Previous studies have focused on the effect of gravity and shown that space conditions allow successful cultivation of Brassica species (Musgrave et al., 2000). However, the quality, biomass, sensitivity of growth to day length, and (constant) temperature are important parameters for this species that have yet to be studied, especially in light of results that seed development in space cultures was compromised (Kuang et al., 2005), especially when oxygen was limited (Kuang et al., 1998). Equally unknown is the effect of environmental conditions in spacecrafts on the performance of plants. Understandably, the emphasis on space biology focuses on the effect of weightlessness and gravity on plant orientation and mechanical properties (Hoson 1999). However, conditions that are the consequence of weightlessness, e.g., lack of convection or altered atmosphere, are equally significant but unknown. Our proposal intends to measure the metabolic, physiological, and genetic responses of radishes grown on the ISS and identify the effect of the space environment on metabolite accumulation, flavor, enzyme activity, mineral uptake, and generation time. This knowledge will be essential for the transition from Earth-bound cultivation of plants to growth conditions in space; the research is important for the establishment of human colonies on the Moon, Mars, and farther outposts. References Hoson T (1999) Interaction of gravity with other environmental factors in growth and development: An introduction. In: Duke PJ, Barlow PW, Hoson T, Mogami Y (eds) Life Sciences: Microgravity Research I, pp 1971-1974 Kuang A, Crispi M, Musgrave ME (1998) Control of seed development in Arabidopsis thaliana by atmospheric oxygen. Plant Cell and Environment 21: 71-78 Kuang A, Popova A, McClure G, Musgrave ME (2005) Dynamics of storage reserve deposition during Brassica rapa L. pollen and seed development in microgravity. Int J Plant Sciences 166: 85-96 Massa G, Newsham G, Hummerick ME, Caro JL, Stutte GW, Morrow RC, Wheeler RM (2013) Preliminary Species and Media Selection for the VEGGIE Space Hardware. Grav Space Res 1: 95-106 Musgrave M, Kuang A, Xiao Y, Stout S, Bingham G, Briarty L, Levinskikh M, Sychev V, Podolski I (2000) Gravity independence of seed-to-seed cycling in Brassica rapa. Planta 210: 400-406 Shen D, Sun H, Huang M, Zheng Y, Li X, Fei Z (2013) RadishBase: A Database for Genomics and Genetics of Radish. Plant Cell Physiol 54: E3-+
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objective to grow plants under space conditions requires optimization of auxiliary parameters such as light intensity, quality, watering, substrate definition, fertilization that all have immediate implications for more efficient agricultural practices. The data that we will obtain examine much more than simple yield (mass) of the plants but include the mineral content, propensity to flower, flavor components, and water use. All attempts will not only optimize plant culture under elevated CO2, limited water, and controlled fertilizer provision, but also investigate the genetic responses of plants to mitigate space (or agricultural) stress. The overall result of the project will therefore benefit our generic understanding of factors that control plant growth and productivity in space and on Earth.
	Progress Report for NASA Grant 80NSSC17K0344 Principal Investigator: Dr. Karl H. Hasenstein Period covered by the report: 7/2021 – 10/2022 Name and address of the recipient's institution: University of Louisiana at Lafayette, Lafayette LA 70504 Award number: 80NSSC17K0344 December 5, 2022 Objectives: Germinate seeds and support plant growth for extended cultivation. Control growth and provide adequate water and fertilization. Assessment of variances of plant growth in space and in ground controls. Crew handling aspects of the NASA Advanced Plant Habitat/APH (planting, sampling & harvesting). The space experiment (APH2): The objective of the space flight experiment was to assess crop plant performance in space and analyze the effect of weightlessness, elevated CO2, specific light conditions, and artificial substrate on plant growth, productivity, and changes in gene expression, which was assessed by RNA-sequencing of the space experiments, NASA Kennedy Space Center (KSC) ground control with elevated CO2, and lab control with ambient CO2. The preparations for the flight experiments were completed and the prepared (seeded) science carrier launched on the Cygnus 14 flight on 10/2/20. The first experiment was initiated on November 3 and completed by 11/30 2020. The second grow-out started 12/1 and was completed 12/30/2020. Sample return occurred in two batches. The first half of material was returned May 2nd and received at the University of Louisiana (UL) on June 2nd. The second half of the material was returned on SpaceX 22 on July 9, 2021. The previous report described space and control growth results; this report focuses on the content analyses and RNA-seq data. Plant productivity: The leaf area per quadrant for the first ISS experiment compared well with previous grow-outs (EVT and ? EVT). The weight determinations for the radish bulbs showed an increasing trend but no statistically different values for the space and ground controls. The ambient conditions at the PI lab (gas-chromatography / GC-UL) showed the largest leaf area. The bulb mass showed a similar trend as the leaf area but was not significantly different between any of the samples. The relationship between elevated CO2 indicated a stronger correlation between leaf area and bulb mass than for the ambient CO2 concentration. Elevated CO2 may lead to greater leaf area but not higher bulb mass. Although without apparent effect on plant productivity, a comparison between the water consumption in space and KSC ground control illustrated the effect of weightlessness on the water distribution in the growth substrate. The difference between the top and bottom moisture sensors was apparent for the ground control but not for space data. The early growth period was characterized by low water consumption. As expected, the water demand increased with leaf growth. Larger fluctuations during the last week indicate strain for the water delivery system. Nutritional values: A major element of the proposed work was an assessment of the nutritional value of space grown plant material. We examined the amount of protein and minerals in leaves and bulbs of radishes under space and ground conditions. Similar to the leaf area and bulb mass data, there were no significant differences between space and ground-grown plants or elevated and ambient CO2 concentration. As had been shown previously, the leaf material had higher protein and mineral content compared with the bulbs. Protein was determined as nitrogen using a conversion factor of 6.25 between N and protein. The mineral content showed higher values for leaves than bulbs and the spectrum of minerals between ground controls and space experiments were not significant. Assessment of photosynthetic efficiency: Instead of using a hand-held fluorimeter to collect spot samples, an indirect measurement was implemented that evaluated the photosynthetic activity by calculating the Normalized Differential Vegetative Index (NDVI), which relies on the difference between leaf reflectivity at 750 nm and 650 nm. NDVI = (R750 - R650)/(R750 + R650). The measurements were possible without touching, and therefore resulted in no damage to the leaves. Since the available infrared (IR) camera only allows images from the rear of the APH, no information from plants in the front of the APH could be obtained. The NDVI data showed a decrease in photosynthetic performance for all conditions and may at least partially be related to the increased sampling of older leaves as the plants matured. Although NDVI assessment does not provide the same information as a specific fluorescence-based system, reduced crew time and the non-invasive nature of reflective measurements justified a lower precision data acquisition. Healthy NDVI

Task Progress:	assessments of photosynthesis results in values = 80 %; it is possible that the steady decline was linked with the appearance of necrotic tissue along the rim of most leaves. qPCR analyses of leaf punch samples: The experimental protocol included sampling of leaf tissue at day 10, 18, and 24 of the 27 day grow-out. These samples were intended to assess the temporal variation of relevant genes in leaf tissue. The samples were collected by astronauts and stored in microfuge tubes at -80°C until processing. The storage time for the ground control and space samples were similar (about six months), but differed between samples from the first and second grow-out on the ISS. The quantified genes included Napin Storage Protein (RSN), Peroxidase Precursor (RPP), Glucosinolate Synthase (GIS), Chlorophyll Binding Protein (CBP), and Raphanus myrosinase (RMA). The expression level of these genes was evaluated against the reference genes Actin-7 and Ubiquitin-1. Although the general profile of these genes is similar, the ground control had the highest myrosinase levels. The transcription was reduced during initial bulb formation (day 18) but recovers by day 24, three days before harvest. In contrast, samples from ISS1 show a steady decline in myrosinase. These data confirm that gene expression does change during the growth period and that leaf punch data can serve the intended purpose, namely evaluating plant health, culture conformity, and comparisons between different growth conditions. Comparison between tissue extraction and Solid Phase Gene Extraction (SPGE): The second ISS grow-out included a test for a simplified genetic sampling method and comparing it with the ground control. SPGE is based on oligo-dT-functionalized probes that upon insertion into tissue extract mRNA based on the hybridization between the oligo-dA tail that is characteristic for mRNA and the oligo-dT layer of the SPGE probe. Although the extraction material is best processed immediately after extraction, preliminary tests showed that the mRNA remains intact for 16 weeks, less than the required storage at -80°C. Despite these uncertainties, SPGE extraction resembled the quantification from leaf punch samples from ISS2 grow-out and ground control. This observation validates SPGE as efficient, fast, and reliable extraction method. RNA-seq evaluation: The overall assessment of differences between space conditions and CO2 effects required three comparisons, a space experiment at predetermined (2500 ppm CO2), a ground control at the same CO2 concentration, and a ground control at ambient (~400 ppm CO2). RNA-seq also allowed for a comparison between the first and second grow-out on the ISS. Sequencing relied on space material (bulb and leaves, 4 replicates each); KSC (= ground control, bulbs, and leaves, 3 replicates each); and UL control (bulbs and leaves, 3 reps each). 28 samples were extracted and sequenced (Novogene, 8801 Folsom Blvd, Sacramento, CA 9582) on an Illumina system resulting in paired end reads of 150 bp. Reference genome and gene model annotation files were obtained from the National Center for Biotechnology Information (NCBI) website:(https://) Alignment to the reference genome was done using Hisat2 v2.0.5. The mapped reads of each sample were assembled by StringTie (v1.3.3b) and full-length transcripts with multiple splice variants quantified for each gene locus with featureCounts (v.1.5.0). Fragments Per Kilobase of transcript sequence per Millions base pairs sequenced (FPKM) of each gene was calculated for each gene. The DESeq2 R-package (1.20.0) was used for determining differential expression. The resulting P-values were adjusted with Benjamini and Hochberg's method. Genes with an adjusted P-value of <=0.05 were assumed to be differentially expressed. Adjusted p-values multiplied by ln(2) transformed expression values were used to obtain a combined, linearized parameter that estimates significant changes between treatments. A correlation analysis verified sample reliability and repeatability of the experiment. A Principal Component Analysis (PCA) was performed on the FPKM values of all samples and showed strong segregation of leaf and bulb data. Separation between treatments clustered leaves into three distinct groups, Space, Ground control with elevated, and ambient CO2. The Correlation and PCA analyses were further corroborated by the examining the overall gene counts that are differentially expressed under different experimental conditions. Differentially expressed gene counts show reduced diversity between the first and second ISS grow-outs and greater variability between leaf material, and under different CO2 conditions Up regulation occurred as often as down regulation. Differentially expressed transcripts of down- and up-regulated genes were considered for pathway analyses if they resulted in more than two-fold difference and a p-value of <0.05. Data of higher stringency (more than five-fold down or up-regulated and p <0.001) were used to identify affected enzymes. The number of altered genes under similar conditions (i.e., first and second ISS grow-out) can further provide a base value of gene variability without space or stress-related responses. This variability is higher in leaves than in bulbs, and by the time of harvest ranges from about 34 genes for bulbs and 158 genes for leaves. Further work will try to identify an associated error range for these estimates. The larger contribution of CO2 effects as primary factor indicates that atmospheric conditions, including airflow, play an important role for plant growth experiments. Future experiments should study different gas concentrations to develop a comprehensive data set on plant susceptibility to pressure, temperature, and partial pressure of included gases. d13C analysis: An attempt was made to measure the isotope fractionation in space-grown plants. d13C‰= [(13C/12C sample)/(13C/12C Std)-1]x1000. We determined the d13C value for ISS (1&2) and ground control samples. The data showed large differences between all conditions and the values deviated from the standard value for C3 plants (-30‰)The shift (likely based on the type of CO2 compression) made the attempted analysis unreliable as the 13C value of the utilized CO2 was not known. Summary: The space experiment APH2 was successfully completed, and most of the analytical work accomplished. The main conclusions are that radishes can grow well in space, have similar productivity as on Earth, and respond to elevated CO2 more than to weightlessness. RNA-seq analyses identify photosynthesis and carbohydrate metabolism as major affected pathways. The rich data sets will be published, and the sequencing data be deposited into NASA's GeneLab database.

Bibliography Type:	Description: (Last Updated: 01/20/2026)
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